CSE217
INTRODUCTION TO DATA SCIENCE

LECTURE 2: EXPLORATORY DATA ANALYSIS

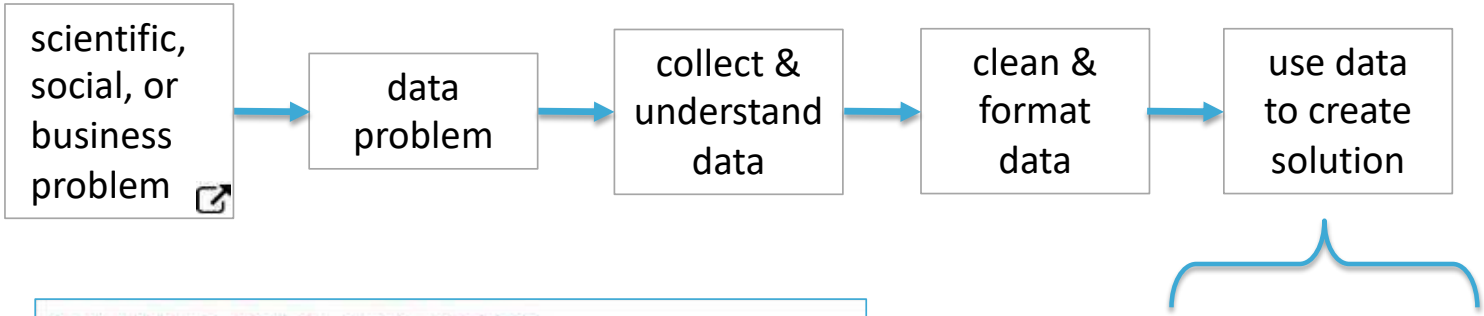
Spring 2019

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RECAP: WHAT IS DATA SCIENCE?

…solving problems with data…



WHAT IS DATA ANALYSIS?

…using data to discover useful information…



- **data**: anything you can *measure* or *record*



- **statistics**: summarize (and visualize) *main characteristics* of the data



- **algorithms**: apply algorithms to find *patterns* in the data

WHERE DOES DATA COME FROM?

- Internal Sources
  - business-centric data in organizational data bases recording day to day operations
  - scientific or experimental data
- Existing External Sources → data is available for free or a fee
  - public government databases, stock market data, Yelp reviews
  - usually (somewhat) pre-processed
- Collect your own data → beyond the scope of this course
- Online Data → typically raw data
  - from APIs (e.g. Google Map API, Facebook API, Twitter API)
  - web scraping: using software, scripts or by-hand extracting data from what is displayed on a page or what is contained in the HTML file

Caution: not all data that is **accessible** is **good to be used**!

- Are you violating their terms of service?
- Privacy concerns for website and their clients?
- Do they have an API or fee that you are bypassing?
- Are they willing to share this data?

VARIABLES AND DATA TYPES

- Types of Variables

① numeric ← order (continuous or discrete)
② categorical ← no order
③ binary (categorical w 2 categories YES/NO)

- Data Types

- integer → discrete, categorical, binary
- Boolean → binary
- floating point → continuous
- string → formatted text
  - → categorical
  - → free-form text
- compound data types (lists, dictionaries, arrays)

We prefer numeric data types & arrays

Example:
https://www.zillow.com

DATA(SET) REPRESENTATION

- Tables (csv, xlsx etc.)
  - two-dimensional representation
    - rows represent data records
    - columns represents one type of measurement
- Structured Data (json, xml etc.)
  - complex and multi-tiered dictionary
- Semi-structured Data (.txt)
  - flat text representation with known structure
  - data can be easily parsed
- Unstructured Data (.txt)
  - prose text

DATA IS (ALWAYS) MESSY

- Common issues with data:
  - missing values: how do we fill in?
  - wrong values: how can we detect and correct?
  - messy format/representation
    - Example: number of produce deliveries over a weekend

	Friday	Saturday	Sunday
Morning	15	158	10
Afternoon	2	90	20
Evening	55	12	45

Common causes of messiness:
- variables/features are stored in both rows and columns
- multiple features are stored in one column
- multiple types of experimental units stored in same table

DATA (PRE-)PROCESSING

Goal: bring data in a format we can use for analysis (and/or machine learning)

→ use a format that is good for Python ☺ (e.g. 2d arrays)

→ recall from last lecture: *data points* vs *features/variables*

- Data Parsing and Formatting
- Data Profiling → asses data amount and quality
- Data Cleaning
- Data Engineering (more later in this course…)
  - detect outliers
  - feature engineering
  - data augmentation

data wrangling

DATA ≠ DATA

- Two kinds of data: *population* vs. *sample*

A *population* is the **entire** set of objects or events under study. Population can be hypothetical "all students" or all students in this class.

A *sample* is a (*representative*) **subset** of the objects or events under study.
→ needed because it's **impossible** or **intractable** to obtain or use population data.

- *What are problems with sample data?*

EXPLORATORY DATA ANALYSIS (EDA)

Different ways of exploring data:

- explore **each individual variable** in the dataset
  - summary statistics
  - spread
  - distribution
- assess **interactions between variables** (or between individual variables and the target)
  - correlation, analysis of variance (ANOVA)
- explore data across many dimensions (more later in this course…)
  - clustering
  - dimensionality reduction (e.g. principal component analysis (PCA), etc.)

SUMMARY STATISTICS

- (sample) mean

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

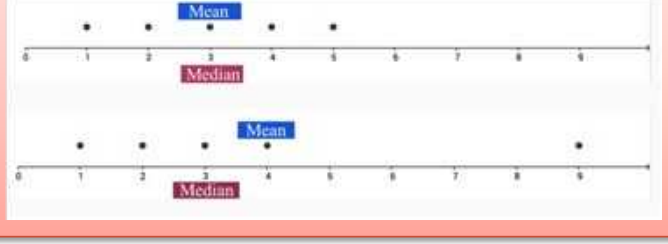
- (sample) median

$$= \begin{cases} x_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{x_{n/2} + x_{(n+1)/2}}{2} & \text{if } n \text{ is even} \end{cases}$$

  - Example: Ages: 17, 19, 21, 22, 23, 23, 23, 38
    What is the median age? What is the mean/average age?

- mean vs median
  - which one is *easier*/more efficient to compute?

Caution: the mean is sensitive to outliers!

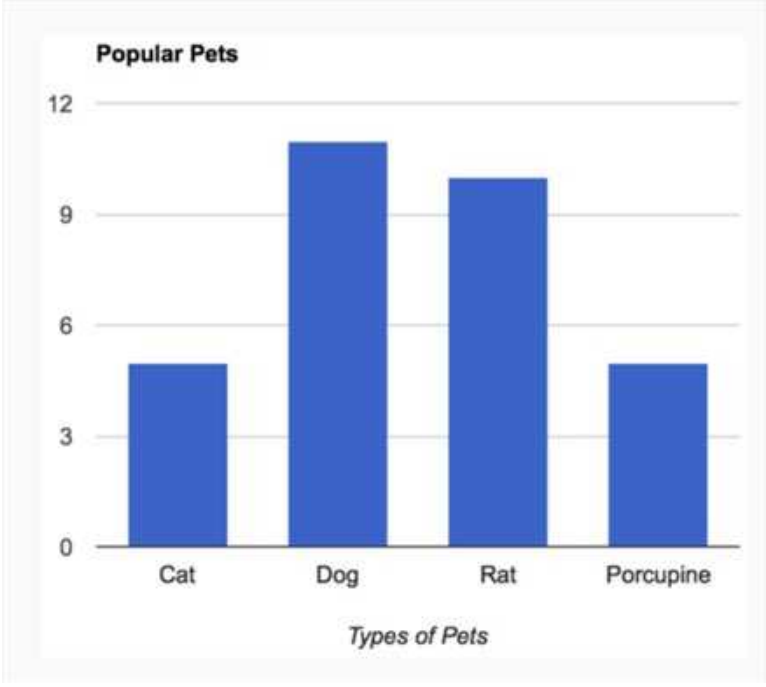


Caution: consider practicality (efficiency) of implementation!

SUMMARY STATISTICS

- mode = variable that occurs most often
  - useful for *categorical* variables

  → visualize with a *bar plot*



DSFS Ch3

MEASURES OF SPREAD

- range = max value – min value
- variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} |x_i - \bar{x}|^2$$

  - Caution: does not have the same unit as x_i
- standard deviation

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} |x_i - \bar{x}|^2}$$

Why is measuring the spread important?

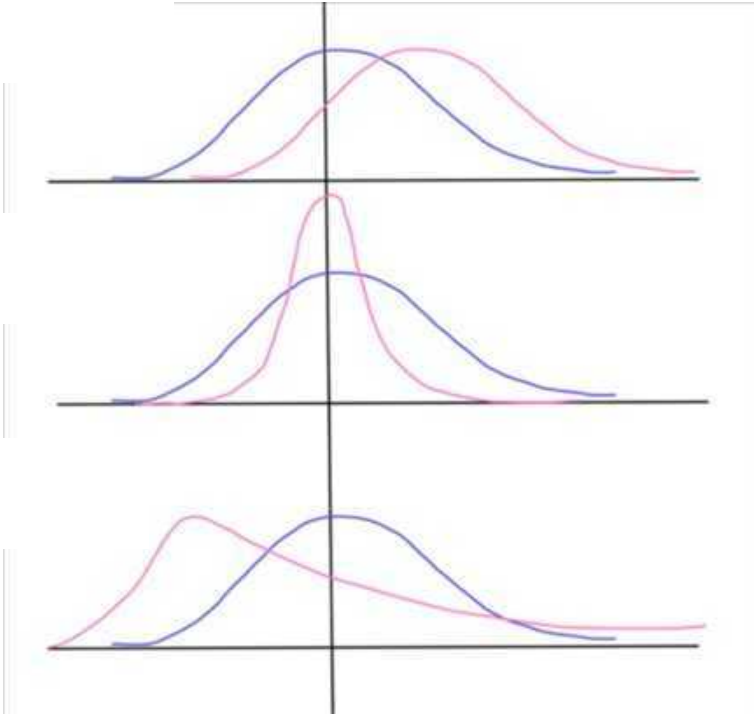
DATA VISUALIZATION

- Can summary statistics and measures of spread tell us everything?

	Dataset I		Dataset II		Dataset III		Dataset IV	
	x	y	x	y	x	y	x	y
	10	8.04	10	9.14	10	7.46	8	6.58
	8	6.95	8	8.14	8	6.77	8	5.76
	13	7.58	13	8.74	13	12.74	8	7.71
	9	8.81	9	8.77	9	7.11	8	8.84
	11	8.33	11	9.26	11	7.81	8	8.47
	14	9.96	14	8.1	14	8.84	8	7.04
	6	7.24	6	6.13	6	6.08	8	5.25
	4	4.26	4	3.1	4	5.39	19	12.5
	12	10.84	12	9.13	12	8.15	8	5.56
	7	4.82	7	7.26	7	6.42	8	7.91
	5	5.68	5	4.74	5	5.73	8	6.89
Sum:	99.00	82.51	99.00	82.51	99.00	82.51	99.00	82.51
Avg:	9.00	7.50	9.00	7.50	9.00	7.50	9.00	7.50
Std:	3.32	2.03	3.32	2.03	3.32	2.03	3.32	2.03

DATA VISUALIZATION

- Can summary statistics and measures of spread tell us everything?



TYPES OF VISUALIZATION

- *distribution*

  → how does a variable distribute over a range of possible values

- *relationship*

  → how do the values of multiple variables in the dataset relate

- *comparison*

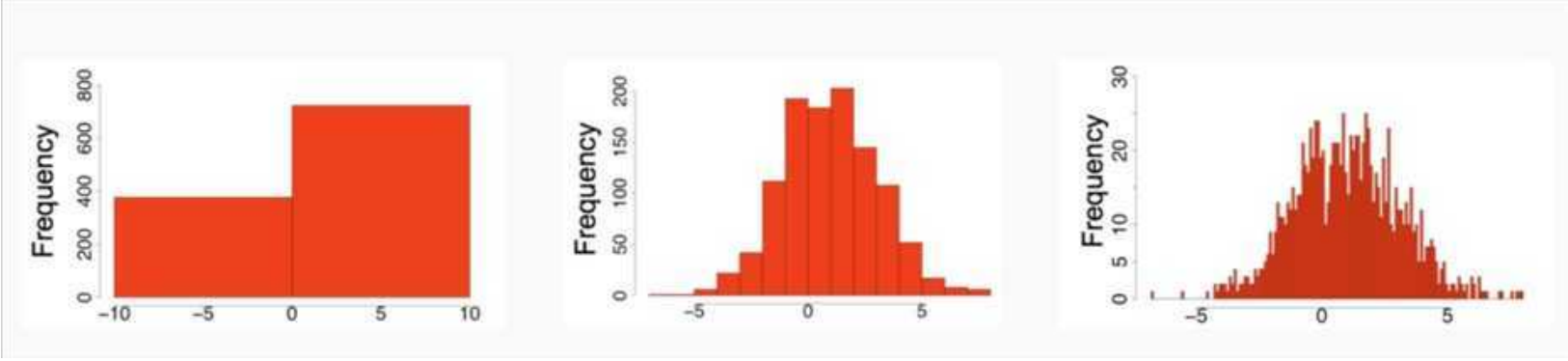
  → how do trends in multiple variable or datasets compare

- *composition*

  → how does the dataset break down into subgroups

VISUALIZE DISTRIBUTION

- *histogram*

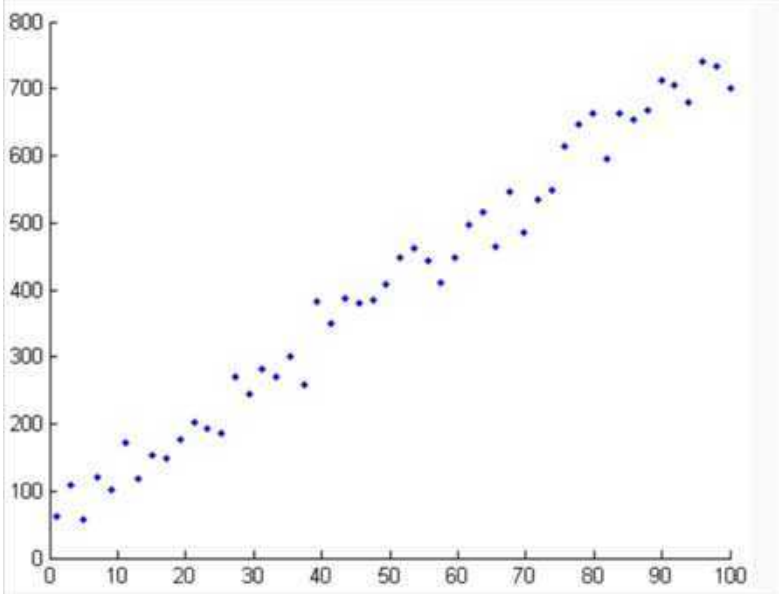


Caution: Trends in histograms are sensitive to the number of bins.

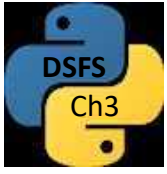
PDSH p245

VISUALIZE RELATIONSHIP

- *scatter plot*
  - distribution of two variables
  - relationship between two variables

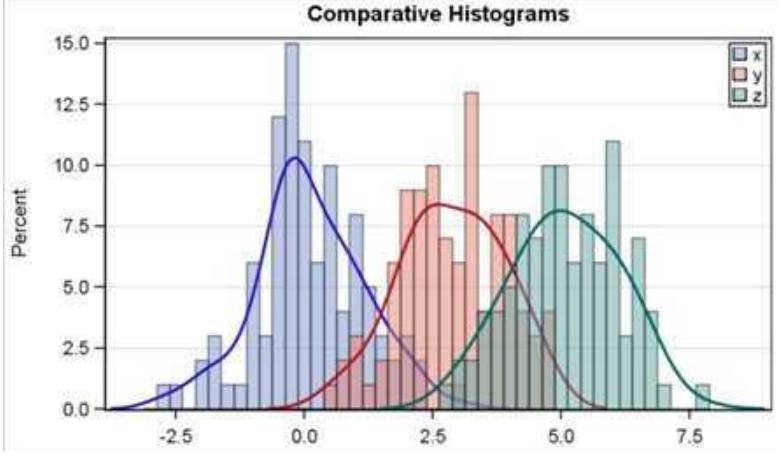






VISUALIZE COMPARISONS

- *multiple histograms*
  - visualize how different variables compare (or how a variable differs over specific groups)

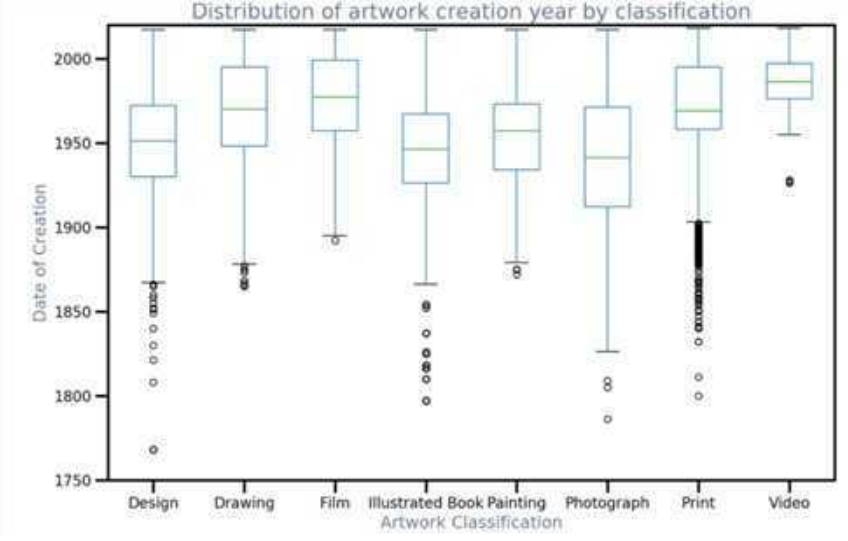


→ we can also use box plots to compare different variables

VISUALIZE COMPOSITION/COMPARISON

- *box plots*
  - **compare** different variables → cf. Lab1
  - compare a quantitative variable **across groups**

  → highlights the *range*, *quartiles*, *median* and *outliers*

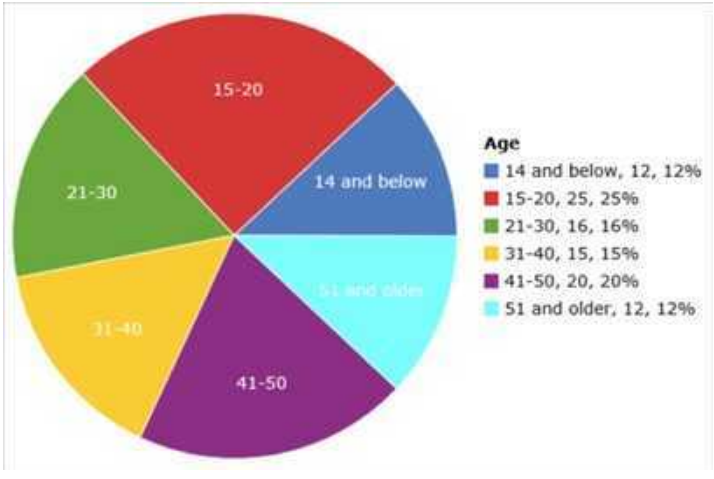


This plot illustrates **composition**, since it looks at classes/categories of <u>one</u> variable.

Lab1

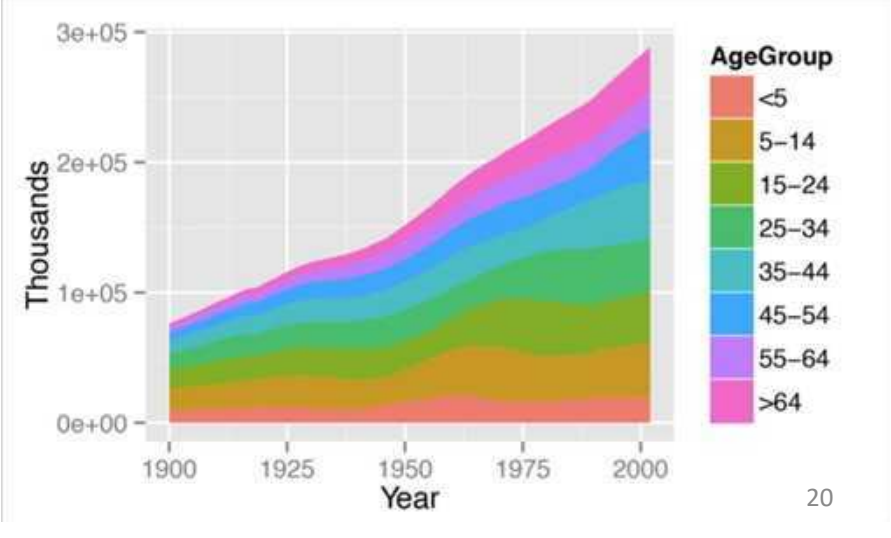
VISUALIZE COMPOSITION

- *pie chart*



- *stacked area graph*

Visualize trend over time!



ACTIVITY 2

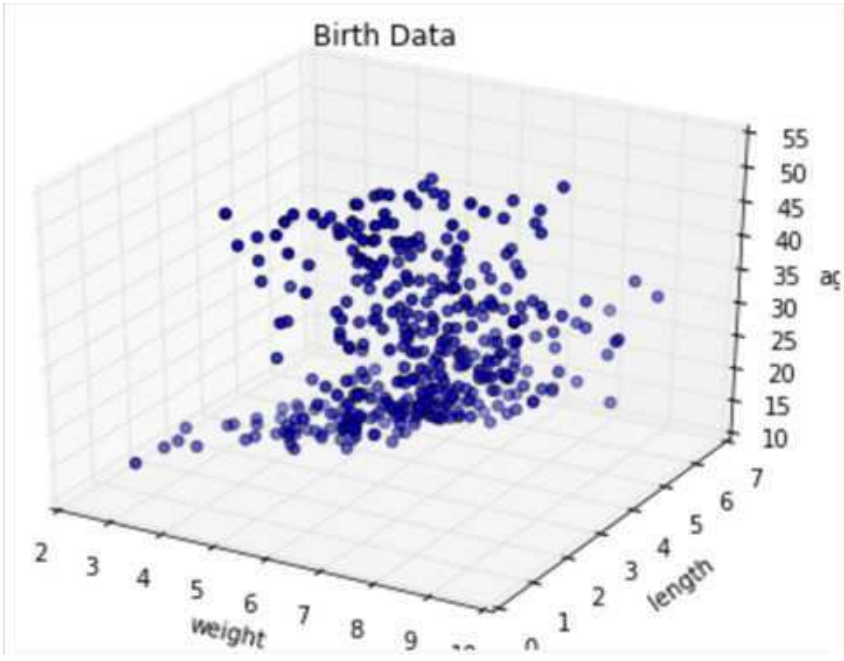
- **TASK 1**: What do the following plots produced in Lab1 visualize?

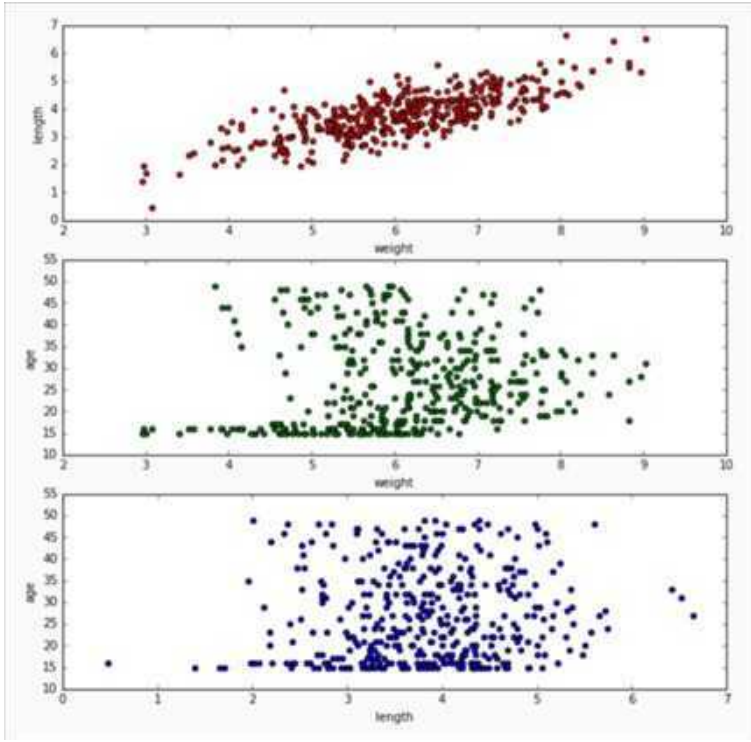
Caution: Not all visualizations are *good* visualizations.

- **TASK 2**: Which of the following visualizations are good/proper visualization and which do you think are problematic (and *why*)?

MORE DIMENSIONS

- How about relationship between 3 variables?

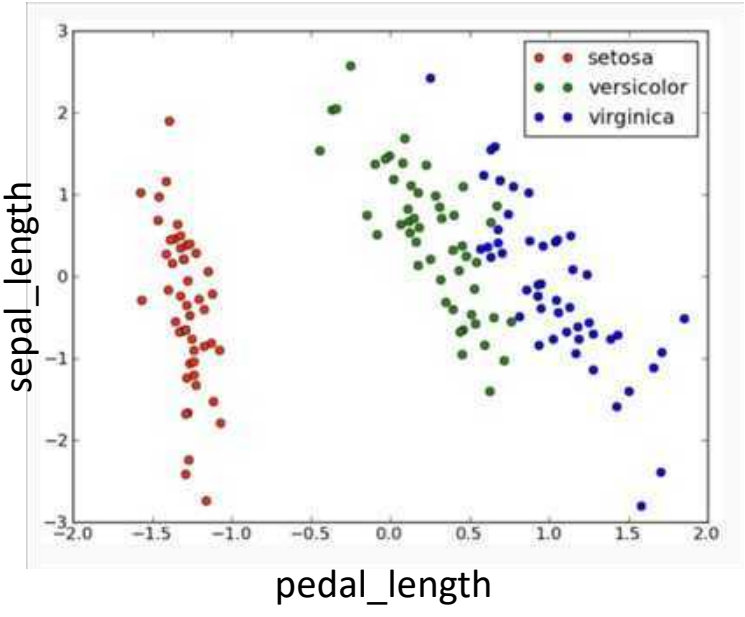




→ 3D is not always better

CATEGORICAL VARIABLES

- use *color coding* for categorical variables



Data visualization can help figure out what we need to **predict** class labels!

SUMMARY & READING

- EDA process
  - (pre-)process data
  - summarize data
  - present/visualize distribution and relationships
- EDA goals
  - develop/find hypothesis/question(s) to be investigated
  - use data to answer the question(s)

- **DSFS**
  - Ch3: Visualizing Data (matplotlib, bar/line charts, scatter plots)
- **PDSH**
  - Ch4: Visualization with Matplotlib
    - plotting with matplotlib (p217-221)
    - scatter plots (p233-237)
    - histograms (p245-247)

